

WASP-39b

R-1656

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_160312 · created 2026-08-02T08:43:21+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457459.8326 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1593 +/- 0.0061
Transit depth (Rp/Rs)^2	2.54 +/- 0.19 [%]
Orbital Inclination (inc)	87.38 +/- 0.39
Airmass coefficient 1 (a1)	1.1475 +/- 0.0096
Airmass coefficient 2 (a2)	-0.1078 +/- 0.0012
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	None
Optimal Aperture	3.79
Optimal Annulus	14.19
Transit Duration (day)	0.1174 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1593 $\pm$ 0.0061 vs 0.1457 (deviation 2.15 σ)
Duration fit vs published	0.1174 d vs 0.12 d (deviation 0.53 σ)
Mid-time O–C	9.7 min
Depth SNR	25.1
Residual scatter	0.82 %

Light curve

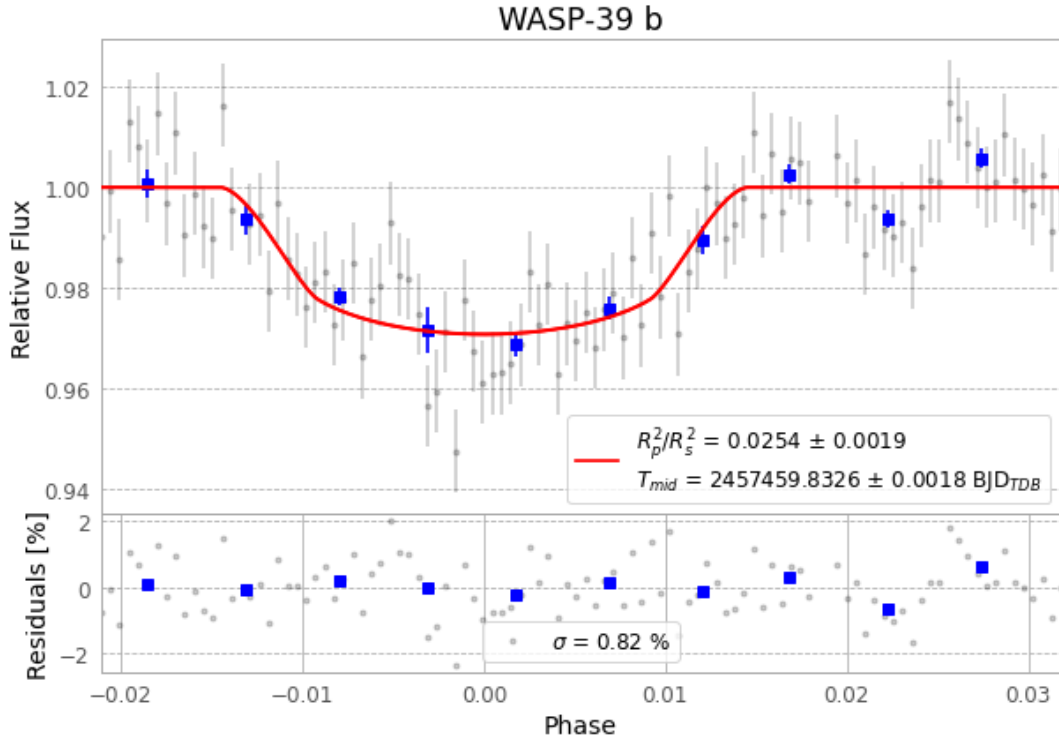


Figure 1 — FinalLightCurve_WASP-39 b_2016-03-11.png (SHA-256 efac0a9332b69894...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [182, 181], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a4b143a9f7e68ebd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

102 FITS frames (67 MB), WASP-39160312054812.FITS → WASP-39160312105712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-160312.log (6d3812eea4a4...), FinalLightCurve_WASP-39 b_2016-03-11.png (efac0a9332b6...), FinalLightCurve_WASP-39 b_2016-03-11.csv (a5ddd193175e...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 08:43Z.